MATH-1730-04
QUESTIONS USED FOR REVIEWS IN CLASS

1. Evaluate the following integrals. Show your work.

(a) $\displaystyle\int \frac{x^2}{\sqrt{1-x}}\,dx.$ (**S.R**: $u = \sqrt{1-x}$ or $u = 1-x$.)

(b) $\displaystyle\int (\ln x)^3 dx.$ (**IBP three times**) Answer: $x((\ln x)^3 - 3(\ln x)^2 + 6\ln x - 6) + C$

(c) $\displaystyle\int x^{11} e^{x^6}\,dx.$ (**S.R** ($u = x^6$) **and IBP**) Answer: $\frac{(x^6-1)e^{x^6}}{6} + C$

(d) $\displaystyle\int \tan^3 x\,\sec x\,dx.$ (**Trig Integral**) Answer: $\frac{\sec^3 x}{3} - \sec x + C$

(e) $\displaystyle\int \frac{e^x}{(e^x-2)(e^{2x}+1)}\,dx.$ (**S.R** ($u = e^x$) **and PF**)

Answer: $\frac{1}{5}\ln|e^x - 2| - \frac{1}{10}\ln(e^{2x}+1) - \frac{2}{5}\tan^{-1}(e^x) + C$

(f) $\displaystyle\int \frac{1}{\sqrt{x^2-4}}\,dx.$ (**Trig Substitution**) Answer: $\ln|\sqrt{x^2-4} + x| + C$

(g) $\displaystyle\int \frac{x+2}{x(x^2-4x+3)}\,dx = \int \frac{x+2}{x(x-1)(x-3)}\,dx.$ (**PF**)

(h) $\displaystyle\int \frac{x}{x^2-1}\,dx.$ (**GPR or S.R** ($u = x^2$))

2. Evaluate the following integrals. Show your work.

(a) $\displaystyle\int_{-\pi/2}^{\pi/2} \frac{x^2 \sin x}{1+x^6}\,dx.$ Answer: 0, since it is an integral of an odd function on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(b) $\displaystyle\int_1^{\infty} \frac{\ln x}{x}\,dx.$ Answer: ∞, divergent

3. Evaluate $\dfrac{d}{dx}\displaystyle\int_5^{x^2+\cos x} \dfrac{\sin t}{t^2+3}\,dt$. $\big($**FTC-Part 1** and **Chain Rule**: $\dfrac{d}{dx}\displaystyle\int_a^{u(x)} f(t)dt = f(u(x))u'(x)\big)$

Answer: $\dfrac{\sin(x^2+\cos x)}{(x^2+\cos x)^2+3}\,(2x-\sin x)$

4. Find the area of the region bounded by the curve $x = y^2 - 4y$ and $x = 2y - y^2$.

Answer: 9

5. Use shell method to find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $y = 0$ and $x = 1$ about the x-axis.

Answer: $\frac{\pi}{5}$

6. Use disk method to find the volume of a sphere of radius R.

Answer: $\frac{4}{3}\pi R^3$

7. Find the surface area obtained by rotating the curve $y = x^3$ $(0 \le x \le 1)$ about the x-axis.

Answer: $\dfrac{\pi}{27}\left(10^{\frac{3}{2}} - 1\right)$.

8. Find the average value of $f(x) = \cos^4 x\,\sin x$ on $[0, \pi]$. Answer: $\frac{2}{5\pi}$.

9. Determine whether the following sequences are convergent or divergent and show your reasoning. If the sequence is convergent, determine its limit.

(a) $a_n = \dfrac{e^n - e^{-n}}{e^n + e^{-n}}$. Answer: $a_n = \dfrac{1-e^{-2n}}{1+e^{-2n}} \to 1$

(b) $b_n = \dfrac{(-1)^n(n^3+4n^2+5)}{n^6+7n^4+n^2+1}$. Answer: 0, since $|b_n| \to 0$.

(c) $c_n = \dfrac{(-1)^n(4n^2+5)}{n^2+1}$. Answer: divergent, since $c_{2n-1} \to -4$ and $c_{2n} \to 4$.

10. Determine whether the series is convergent or divergent. Show your reasoning and indicate the test that is used in your solution. If it is convergent, specify whether it is absolutely convergent or conditionally convergent.

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{n^2}{n^2-1}$. (**Test for Divergence**) Answer: divergent $(a_n \to 1 \neq 0)$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$. (**Comparison Test**) Answer: convergent $\left(\dfrac{a_n}{1/n^2} \to \dfrac{5}{2}\right)$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{n+5}{5^n}$. (**Ratio/Root Test**) Answer: convergent $\left(\frac{a_{n+1}}{a_n} \to \frac{1}{5} \text{ or } \sqrt[n]{a_n} \to \frac{1}{5}\right)$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{n^{3n}}{(n!)^n}$. (**Ratio/Root/Comparison Test**) Answer: convergent

$\left(\frac{a_{n+1}}{a_n} \to 0, \text{ or } \sqrt[n]{a_n} \to 0, \text{ or } a_n = \left(\frac{n^3}{n!}\right)^n \leq \frac{1}{2^n} \text{ when } n \geq 7\right)$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(4n)}{4^n}$. (**Comparison Test**) Answer: Absolutely convergent $(|a_n| \leq \frac{1}{4^n})$

(f) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^3+5}$. (**Alternating Series and Comparison Tests**)

Answer: conditionally convergent $\left(\frac{|a_n|}{1/n} \to 1\right)$

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n + 3^n}{1+4^n}$. (**Ratio/Comparison Test**) Answer: convergent

$\left(\frac{a_{n+1}}{a_n} \to \frac{3}{4}, \text{ or } \frac{a_n}{(3/4)^n} \to 1\right)$

(h) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$. (**Integral Test**) Answer: divergent $\left(\displaystyle\int_2^{\infty} \frac{1}{x \ln x}\, dx = \infty\right)$

11. Use **definition** to find the MacLaurin series of the function $f(x) = \ln(1+5x)$, and determine the radius of convergence. Answer: $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}5^n}{n} x^n,\ R = \tfrac{1}{5}.$

Now $f(0) = 0,\ f^{(n)}(x) = (-1)^{n-1}\, 5^n\, (n-1)!\, (1+5x)^{-n}$ and $f^{(n)}(0) = (-1)^{n-1}\, 5^n\, (n-1)!$

12. For each of the following power series, find the radius of convergence and the interval of convergence. Show your reasoning and indicate the test that is used in your solution.

(a) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!}.$ Answer: $R = \infty,\ I = (-\infty, \infty)$

(b) $\displaystyle\sum_{n=1}^{\infty} x^n n^n.$ Answer: $R = 0,\ I = [0, 0]$

(c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n+3}}{6^n} (x+5)^n.$ Answer: $R = 6,\ I = (-11, 1)$

Test for Divergence is used for $x = -11$ and $x = 1$.

13. Let $f(x) = \tan^{-1} x$.

(a) Find a power series representation for $f'(x)$. Show your work.

(b) Use the power series in (a) to find a power series representation for $f(x)$.

Answer: see Example 2(3) in the lecture outline of Section 11.9.

14. Evaluate $\displaystyle\int \cos(x^2)\, dx$ and $\displaystyle\int_0^1 \cos(x^2)\, dx$ as infinite series. Show your work.

(**Use the MacLaurin series of** $\cos(x)$ **and then the FTC - Part 2**.)

15. Find the sum for each of the following series. Show your work.

(a) $\displaystyle\sum_{n=0}^{\infty}(-1)^n \frac{(\pi/6)^{2n+1}}{(2n+1)!}$. (**Use the MacLaurin series of** $\sin x$) Answer: $\frac{1}{2}$

(b) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n}{n!}$. (**Use the MacLaurin series of** e^x) Answer: $e^{-1}-1$

(c) $\displaystyle\sum_{n=0}^{\infty}\frac{2^{3n+1}}{3^{2n+1}}$. (**Geometric series**) Answer: 6

$$\sum_{n=0}^{\infty}\frac{2^{3n+1}}{3^{2n+1}} = \sum_{n=0}^{\infty}\frac{2}{3}\left(\frac{8}{9}\right)^n = \frac{2/3}{1 - 8/9} = 6$$

(d) $\displaystyle\sum_{n=1}^{\infty}\frac{3}{n(n+3)}$. (**Telescoping series**) Answer: $\frac{11}{6}$

$$\sum_{n=1}^{\infty}\frac{3}{n(n+3)} = \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+3}\right) = \frac{11}{6}$$

16. Find a power series representation for each of the following functions. Also find the radius of convergence and the interval of convergence. Show your work.

(a) $f(x) = \dfrac{x}{1+2x^2}$.

Answer: $f(x) = x\dfrac{1}{1-(-2x^2)} = \displaystyle\sum_{n=0}^{\infty}(-2)^n x^{2n+1}$, $R = \frac{1}{\sqrt{2}}$, $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(b) $f(x) = \dfrac{1+x}{1-x}$.

Answer: $f(x) = -1 + \dfrac{2}{1-x} = 1 + \displaystyle\sum_{n=1}^{\infty} 2x^n$, $R = 1$, $(-1, 1)$